7.3

Integrals of the form

$$\int \sin mx \cos nx\, dx, \quad \int \sin mx \sin nx\, dx, \quad \int \cos mx \cos nx\, dx$$

can be found by using the trigonometric identities

$$\sin\alpha\cos\beta = \tfrac{1}{2}[\sin(\alpha-\beta) + \sin(\alpha+\beta)]$$

$$\sin\alpha\sin\beta = \tfrac{1}{2}[\cos(\alpha-\beta) - \cos(\alpha+\beta)]$$

$$\cos\alpha\cos\beta = \tfrac{1}{2}[\cos(\alpha-\beta) + \cos(\alpha+\beta)]$$

Example: $I = \int \sin x \cdot \sin 2x \, dx$

$$I = \int \frac{1}{2}\left[\cos(x-2x) - \cos(x+2x)\right]dx$$

$$= \int \frac{1}{2}\left[\cos(-x) - \cos 3x\right]dx$$

note: $\sin(-x) = -\sin x$, $\cos(-x) = \cos x$

$$= \int \frac{1}{2}\left[\cos x - \cos 3x\right]dx$$

$$= -\frac{1}{2}\int \cos x\, dx - \frac{1}{2}\int \cos 3x\, dx$$

Formula : $\int \sin mx \, dx = -\frac{1}{m} \cdot \cos mx + C$

$$\int \cos mx \, dx = \frac{1}{m} \sin mx + C$$

$$\Rightarrow I = +\frac{1}{2} \sin x - \frac{1}{2} \cdot \frac{1}{3} \sin 3x + C$$

$$= \frac{1}{2} \sin x - \frac{1}{6} \sin 3x + C$$

Example : $\int 4 \sin x \cdot \cos 2x \cdot \sin 3x \, dx$

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can be found by using the trigonometric identities

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$$\sin \alpha \sin \beta = \tfrac{1}{2}[\cos(\alpha - \beta) - \cos(\alpha + \beta)]$$

$$\cos \alpha \cos \beta = \tfrac{1}{2}[\cos(\alpha - \beta) + \cos(\alpha + \beta)]$$

$$= \int 4 \cdot \frac{1}{2}\left[\sin(x - 2x) + \sin(x + 2x)\right] \cdot \sin 3x \, dx$$

$$= \int 2 \cdot \left[\sin(-x) + \sin 3x\right] \cdot \sin 3x \, dx$$

$$= \int \left[2 \sin(-x) \cdot \sin 3x + 2 \cdot \sin 3x \cdot \sin 3x \right] dx$$

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$$\cos\alpha \cos\beta = \tfrac{1}{2}[\cos(\alpha-\beta) + \cos(\alpha+\beta)]$$

$$= \int \left[\cos(-x-3x) - \cos(-x+3x) + \cos(3x-3x) + \cos(3x+3x) \right] dx$$

$$= \int (\cos 4x - \cos 2x + \cos 0 + \cos 6x)\, dx$$

$$= \frac{\sin 4x}{4} - \frac{\sin 2x}{2} + x + \frac{\sin 6x}{6} + C$$

So far:

① $\int \sin^m x \, dx$, $\int \cos^m x \, dx$

② $\int \sin^m x \cdot \cos^n x \, dx$

③ $\int \sin mx \cdot \cos nx \, dx$

① $\int \sin^5 x \, dx$

② $\int \sin^5 x \cdot \cos^3 x \, dx$

③ $\int 2 \cdot \cos x \cdot \cos 5x \, dx$

$$\int \sin^n x \, dx = -\frac{1}{n} \sin^{n-1} x \cos x + \frac{n-1}{n} \int \sin^{n-2} x \, dx$$

$$\int \cos^n x \, dx = \frac{1}{n} \cos^{n-1} x \sin x + \frac{n-1}{n} \int \cos^{n-2} x \, dx$$

$I = \int \sin^n kx \, dx$

$u = kx \Rightarrow du = k \cdot dx$

$I = \int \sin^n u \cdot \frac{du}{k} = \frac{1}{k} \int \sin^n u \, du$

$I = \frac{1}{k} \left[-\frac{1}{n} \cdot \sin^{n-1} u \cdot \cos u + \frac{n-1}{n} \int \sin^{n-2} u \, du \right]$

$$\int \sin^n kx\,dx = -\frac{1}{kn}\sin^{n-1} kx \cdot \cos kx + \frac{n-1}{n}\int \sin^{n-2} kx\,dx$$